HARD · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

Geometry and Trigonometry

03

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. 18

Choice A is correct. It's given that WZ and XY intersect at point Q. It follows that $\angle WQX$ and $\angle YQZ$ are vertical angles, which means they are congruent to each other. The figure shows that $\angle W$ and $\angle Y$ both have a measure of $a°$, so they are congruent to each other. Therefore, $\triangle QWX$ and $\triangle QYZ$ are similar triangles, where WQ corresponds to YQ and WX corresponds to YZ. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{YQ} = \frac{WX}{WQ}$. It's given that $YQ = 21$, $WQ = 70$, and $WX = 60$. Substituting 21 for YQ, 70 for WQ, and 60 for WX in the equation $\frac{YZ}{YQ} = \frac{WX}{WQ}$ yields $\frac{YZ}{21} = \frac{60}{70}$, or $\frac{YZ}{21} = \frac{6}{7}$. Multiplying each side of this equation by 21 yields $YZ = 18$. Therefore, the length of YZ is 18.

Choice B is incorrect. This is the length of QZ, not YZ.

Choice C is incorrect. This is the length of XQ, not YZ.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2 2592/5

Accept also: `518.4`

The correct answer is 518.4. It's given that the area of the original poster is 360 square inches. Let l represent the length, in inches, of the original poster, and let w represent the width, in inches, of the original poster. Since the area of a rectangle is equal to its length times its width, it follows that $360 = lw$. It's also given that a copy of the poster is made in which the length and width of the original poster are each increased by 20%. It follows that the length of the copy is the length of the original poster plus 20% of the length of the original poster, which is equivalent to $l + \frac{20}{100}l$ inches. This length can be rewritten as $l + 0.2l$ inches, or $1.2l$ inches. Similarly, the width of the copy is the width of the original poster plus 20% of the width of the original poster, which is equivalent to $w + \frac{20}{100}w$ inches. This width can be rewritten as $w + 0.2w$ inches, or $1.2w$ inches. Since the area of a rectangle is equal to its length times its width, it follows that the area, in square inches, of the copy is equal to $1.2l1.2w$, which can be rewritten as $1.21.2lw$. Since $360 = lw$, the area, in square inches, of the copy can be found by substituting 360 for lw in the expression $1.21.2lw$, which yields $1.21.2360$, or 518.4. Therefore, the area of the copy, in square inches, is 518.4.

Q3 5

The correct answer is 5. The standard form of an equation of a circle in the xy-plane is $x - h^2 + y - k^2 = r^2$, where h, k, and r are constants, the coordinates of the center of the circle are h, k, and the length of the radius of the circle is r. It's given that an equation of the circle is $x - 2^2 + y - 9^2 = r^2$. Therefore, the center of this circle is $2, 9$. It's given that the endpoints of a diameter of the circle are $2, 4$ and $2, 14$. The length of the radius is the distance from the center of the circle to an endpoint of a diameter of the circle, which can be found using the distance formula, $\sqrt{x_1 - x_2{}^2 + y_1 - y_2{}^2}$. Substituting the center of the circle $2, 9$ and one endpoint of the diameter $2, 4$ in this formula gives a distance of $\sqrt{2 - 2^2 + 9 - 4^2}$, or $\sqrt{0^2 + 5^2}$, which is equivalent to 5. Since the distance from the center of the circle to an endpoint of a diameter is 5, the value of r is 5.

Q4

.3928

Accept also: `.3929`, `11/28`

The correct answer is $\frac{11}{28}$. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to the angle with measure $x°$ is 11 units and the length of the hypotenuse is 28 units. Therefore, the value of $\cos x°$ is $\frac{11}{28}$. Note that 11/28, .3928, .3929, 0.392, and 0.393 are examples of ways to enter a correct answer.

Q5

C

C. $20°$

Choice C is correct. It's given that triangle XYZ is similar to triangle RST, such that X, Y, and Z correspond to R, S, and T, respectively. Since corresponding angles of similar triangles are congruent, it follows that the measure of $\angle Z$ is congruent to the measure of $\angle T$. It's given that the measure of $\angle Z$ is $20°$. Therefore, the measure of $\angle T$ is $20°$.

Choice A is incorrect and may result from a conceptual error.

Choice B is incorrect. This is half the measure of $\angle Z$.

Choice D is incorrect. This is twice the measure of $\angle Z$.

Q6

D

D. 1,849

Choice D is correct. The area of a circle can be found by using the formula $A = \pi r^2$, where A is the area and r is the radius of the circle. It's given that the radius of circle A is $3n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi 3n^2$, or $9\pi n^2$. It's also given that the radius of circle B is $129n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi 129n^2$, or $16{,}641\pi n^2$. Dividing the area of circle B by the area of circle A gives $\frac{16{,}641\pi n^2}{9\pi n^2}$, which simplifies to $1{,}849$. Therefore, the area of circle B is $1{,}849$ times the area of circle A.

Choice A is incorrect. This is how many times greater the radius of circle B is than the radius of circle A.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the coefficient on the term that describes the radius of circle B.

Q7

D

D. 25π

Choice D is correct. It's given that the circle is a unit circle, which means the circle has a radius of 1. It's also given that point G is the center of the circle and has coordinates $(0,0)$ and that point H lies on the circle and has coordinates $(-1,y)$. Since the radius of the circle is 1, the value of y must be 0, as all other points with an x-coordinate of -1 are a distance greater than 1 away from point G. Since F and H are points on the unit circle centered at G, let side FG be the starting side of the angle and side GH be the terminal side of the angle. Since side FG is on the positive x-axis and side GH is on the negative x-axis, side FG is half of a rotation around the unit circle, or π radians, away from side GH. Therefore, the positive measure of angle FGH, in radians, is equal to π plus an integer multiple of 2π. In other words, the positive measure of angle FGH, in radians, is an odd integer multiple of π. Of the given choices, only 25π is an odd integer multiple of π.

Choice A is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0,-1)$, not $(-1,0)$.

Choice B is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0,1)$, not $(-1,0)$.

Choice C is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(1,0)$, not $(-1,0)$.

Q8

A

A. 5

Choice A is correct. It's given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other angle, the angles are complementary. It follows that $\angle Q$ and $\angle R$ are complementary. That is, the sum of the measures of the angles is 90 degrees. It's given that the measure of $\angle Q$ is $x + 61$ degrees and the measure of $\angle R$ is $4x + 4$ degrees. It follows that $(x + 61) + (4x + 4) = 90$. By combining like terms, this equation can be rewritten as $5x + 65 = 90$. Subtracting 65 from each side of this equation yields $5x = 25$. Dividing each side of this equation by 5 yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\cos(Q) = \cos(R)$ rather than $\cos(Q) = \sin(R)$.

Choice C is incorrect. This would be the value of x if $\cos(Q) = -\cos(R)$ rather than $\cos(Q) = \sin(R)$ and if $\angle R$ were obtuse rather than acute.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

1660

The correct answer is $1{,}660$. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. When two parallel lines are intersected by a transversal line, the angles formed have the following properties: two adjacent angles are supplementary, and alternate interior angles are congruent. Therefore, each of the four acute angles have the same measure, and each of the four obtuse angles have the same measure. It's also given that the measure of one of the acute angles is $9x - 560°$. If two angles are supplementary, then the sum of their measures is $180°$. Therefore, the measure of the obtuse angle adjacent to any of the acute angles is $180 - (9x - 560)°$, or $180 - 9x + 560°$, which is equivalent to $-9x + 740°$. It's given that the sum of the measures of one of the acute angles and three of the obtuse angles is $-18x + w°$. It follows that $(9x - 560) + 3(-9x + 740) = -18x + w$, which is equivalent to $9x - 560 - 27x + 2{,}220 = -18x + w$, or $-18x + 1{,}660 = -18x + w$. Adding $18x$ to both sides of this equation yields $1{,}660 = w$.

Q10 A

A. 6

Choice A is correct. It's given that the length of each side of a scale model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of a ballroom. Therefore, the area of the scale model is $\frac{1}{10}^2$, or $\frac{1}{100}$, times the area of the actual floor of the ballroom. It's given that the area of the floor of the ballroom is 600 square meters. Therefore, the area, in square meters, of the scale model is $\frac{1}{100}600$, or 6.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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